Scalability & Future Plan

Date	4 July 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Uses a static crop dataset	Limits prediction to available data	High
2	No real-time weather integration	Recommendations may not reflect current weather	Medium
3	Runs only on a local server	Cannot be accessed remotely by users	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports local users	Deploy on cloud to support multiple users simultaneously
2	Data Storage	CSV dataset	Use MySQL or MongoDB for larger datasets
3	Performance	Local Flask application	Optimize model and deploy using cloud services
4	Security	Basic input validation	Add user authentication and secure HTTPS communication

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Integrate real-time weather API and fertilizer recommendation	Next 3 Months	More accurate crop recommendations
Phase 3	Deploy the application on a cloud platform and add user authentication	Next 6 Months	Improved accessibility and security
Phase 4	Develop a mobile application with multilingual support	Next 12 Months	Increased usability for farmers across different regions